



Croatian Open Competition in Informatics

Round 1, November 5th 2022

Tasks

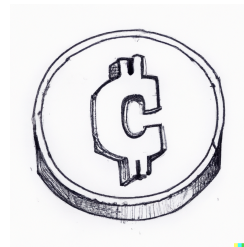
Task	Time limit	Memory limit	Score
Desni klik	1 second	512 MiB	50
Čokolade	1 second	512 MiB	70
Berilij	1 second	512 MiB	110
Iksevi	3 seconds	512 MiB	110
Neboderi	2.5 seconds	512 MiB	110
Total			450



Task Desni klik

NFP is the future!, this is something all Noa's friends can expect him to say when finance topics come up.

NFP is one of the cryptocurrencies. The value of NFP over the course of s days can be represented with a matrix with r rows and s columns, consisting only of characters `.` and `#`. The character `#` in the i -th column represents the value of NFP on the i -th day, with the value being the number of the row, counted bottom-up.



```
....##.  
#...#...  
.##....  
.....#
```

*The value of NFP from the second example over the course of 7 days was: 3, 2, 2, 3, 4, 4, 1.
The insecurity of this NFP is 3.*

The *insecurity* of NFP is defined as the difference between the maximum and minimum value it achieves over the course of s days.

Noa wants to determine the *insecurity* for n NFPs, whose values are represented by matrices with r rows and s columns.

Help him determine the *insecurity* of each of the n NFPs.

Input

The first line contains integers n , r and s ($1 \leq n \leq 20, 2 \leq r, s \leq 50$), the number of NFPs, and the number of rows and columns of the matrices.

n matrices follows, one below another, each with r rows and s columns, representing NFP values. Each column consists only of characters `.`, except for exactly one character `#`.

Output

Print n lines. In the i -th of n lines print the *insecurity* of the i -th NFP.

Scoring

Subtask	Points	Constraints
1	5	$r = s = 2$
2	15	$n = 1$
3	30	No additional constraints.



Examples

input

4 2 2

##

..

..

##

#.

.#

.#

#.

output

0

0

1

1

input

1 5 8

.....#.#

...#...#.

..#.#...

.#.....

#.....

output

4

input

2 3 3

...

##.

..#

.#.

#..

..#

output

1

2

Note: For clarity's sake, in the examples there are blank lines between matrices. In the test cases there will **not** be blank lines between matrices.

Clarification of the first example:

The values of the first and the second NFP do not change over the days, so their insecurities are equal to 0. The value of the third NFP decreases by 1 on the second day, so the insecurity is equal to 1. The value of the fourth NFP increases by 1 on the second day, so the insecurity is equal to 1.

Clarification of the second example:

The maximum value NFP has is 5 (on days 6 and 8), and the minimum value is 1 (on day 1). Therefore, the insecurity is equal to $5 - 1 = 4$.



Task Čokolade

Little Lana and little Fran are visiting a chocolate factory. They have seen how chocolate is made, tasted many chocolates, and now they want to buy some of the chocolates.

In the shop, there are n different chocolates, and the i -th of them has the price c_i . Lana and Fran want to buy m chocolates.

Fran found a way to split the cost in the shop:

- If the chocolate is cheaper than k kunas, Lana will pay for it.
- Otherwise, Lana will pay k kunas, and Fran will pay the rest, that is $c_i - k$ kunas.

Let's denote l as the amount Lana has to pay, and f as the amount Fran has to pay. Lana, dissatisfied with Fran's deal, wants to spite Fran and choose the chocolates so the value of the expression $l - f$ is as small as possible. Since Fran is hesitant and doesn't know how many he wants to buy, Lana wants to know the minimal value of the expression $l - f$ for q different numbers k_i and m_i .

Help her choose the chocolates and determine the minimum value of the expression $l - f$ for each of the q queries.

Input

The first line contains two integers n and q ($1 \leq n, q \leq 10^5$), the number of chocolates, and the number of queries.

The second line contains n integers $c_1, c_2, \dots, c_n$ ($1 \leq c_i \leq 10^9$), the prices of the individual chocolates, in order.

The following q lines contain integers k_i and m_i ($1 \leq k_i \leq 10^9, 1 \leq m_i \leq n$), Fran's bound, and the number of chocolates they are going to buy.

Output

Print q lines. In the i -th line print the answer to Lana's i -th query.

Scoring

Subtask	Points	Constraints
1	15	$n, q \leq 1000, c_i, k_i \leq 10^6$
2	20	$k_1 = \dots = k_n$
3	35	No additional constraints.





Examples

input

5 2
1 9 22 10 19
18 4
5 2

output

34
-21

input

7 4
1 5 4 3 7 11 9
5 4
5 7
7 3
4 5

output

4
16
7
1

input

3 3
5 6 7
10 1
5 3
3 3

output

5
12
0

Clarification of the first example:

In the first query, Lana can take chocolates with prices 1, 9, 22, and 10. Lana will pay 38 kunas, and Fran 4 kunas. The answer is $38 - 4 = 34$.

In the second query, Lana will choose chocolates with prices 22 and 19. She will pay 10 kunas, and Fran will pay 31 kunas. The answer is $10 - 31 = -21$.

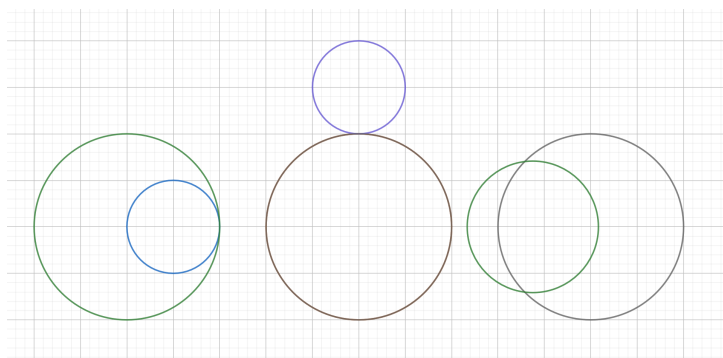
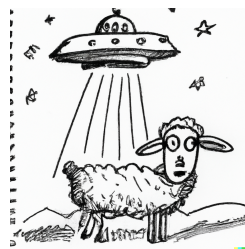


Task Berilij

Little sheep Be (short for Berilij) was kidnapped by the aliens, and they have a rather unusual request for her. They want to hire her.

Precisely on Saturday, November 5th, the aliens plan to visit Earth with n spaceships and reward the best COCI contestants (and maybe hire them too). Their spaceships are perfect circles.

Due to safety reasons, they chose m pairs of spaceships that have to touch externally when they land. They have already determined the landing coordinates of the center point for each of the spaceships, and Be's task is to determine the radius of each of the spaceships, so the conditions are satisfied.



*In the image, the left and the right pairs of spaceships do **not** satisfy the condition of touching externally. The pair of spaceships in the middle fulfills the condition of touching externally.*

Spaceships are very expensive, and their cost is equal to their area, so the aliens are asking Be to determine the radii with the **minimum cost** of the spaceships.

Their advanced technology allows spaceships to overlap, and, even more interestingly, they know how to make a spaceship with radius equal to 0.

If there is no set of radii that satisfies the conditions, aliens expect Be to inform them about it. If Be doesn't succeed in determining the radii, they will hire her as lunch.

Input

The first line contains two integers n and m ($1 \leq n, m \leq 10^5$), the number of spaceships, and the number of conditions.

The following n lines contain real numbers x_i and y_i ($-10\,000 \leq x_i, y_i \leq 10\,000$), the coordinates of the center point of the i -th spaceship. Each of the numbers will be given with 10 decimal places.

The following m lines contain two integers a_i and b_i ($1 \leq a_i, b_i \leq n, a_i \neq b_i$), representing the condition that the a_i -th and b_i -th spaceship must touch externally after landing. For each unordered pair (a_i, b_i) there will be at most one such condition.

Output

If there is no solution, in the first and only line print **NE**. Otherwise, in the first line print **DA**, and in the i -th of the following n lines print the radius of the i -th spaceship.



Scoring

Subtask	Points	Constraints
1	15	n is odd, and each of the spaceships should touch exactly two other spaceships.
2	25	There exists at least one way to determine the radii so the conditions are satisfied.
3	30	For each pair of spaceships (a, b) , there's at most one sequence of distinct spaceships that starts in a , and finishes in b , and all adjacent spaceships in the sequence are necessarily touching.
4	40	No additional constraints.

Your answer will be considered correct if, for each radius of the n spaceships, its absolute or relative error doesn't exceed 10^{-4} . In other words, if your answer for the i -th spaceship is r_{si} , and the correct answer r_{ci} , then your answer will be considered correct if $|r_{si} - r_{ci}| \leq 10^{-4}$ or $|\frac{r_{si} - r_{ci}}{r_{ci}}| \leq 10^{-4}$

Examples

input

```
3 3
0.0000000000 0.0000000000
0.0000000000 2.0000000000
2.0000000000 0.0000000000
1 2
2 3
3 1
```

output

```
DA
0.585786
1.414214
1.414214
```

input

```
5 4
-0.4585133080 0.2893567973
9.9368007273 7.1806641913
-8.4621834970 -2.8309311865
0.0122121945 -2.8309311865
2.3991780589 -8.8626906628
```

2 1

3 2

4 3

5 1

output

```
DA
0.000000
12.472076
8.474396
0.000000
9.587824
```

input

```
5 5
0.0000000000 0.0000000000
1.0000000000 2.0000000000
2.0000000000 4.0000000000
3.0000000000 6.0000000000
4.0000000000 8.0000000000
```

1 2

2 3

3 4

4 5

5 1

output

NE

Clarification of the first example:

This is the only solution that satisfies all the touching conditions. Please note that solution (0.585700, 1.414357, 1.414357) is also considered correct, even though spaceships 2 and 3 aren't touching, because the absolute error doesn't exceed 10^{-4} .

Clarification of the third example:

There is no arrangement of the radii that satisfies all conditions.



Task Iksevi

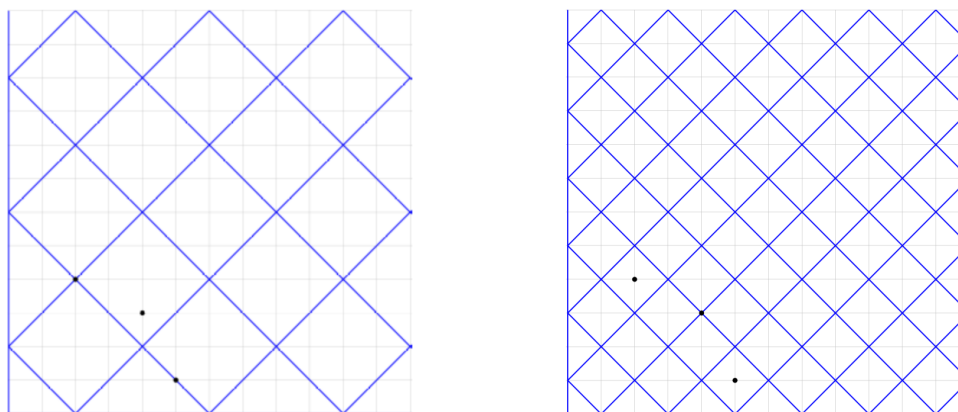
After ten years of programming, Vinko has decided to change his profession and become a ceramist. Already on the first day of his new job he got an extremely difficult task.

He has to pave the concert hall's floor with square ceramic tiles. However, he won't pave the floor so that the tiles are parallel with the hall's walls. Instead, he will rotate them so that the diagonals of the tiles are parallel to the walls.



Vinko has not decided which size of tiles he will use, but he knows they all have to be the same size, and that the length of their diagonals in millimeters has to be a positive even integer. He will put the first tile so that it touches the bottom and left walls, and then he will pave the others so that they share a side with some of the previously set tiles. He will repeat the procedure until he paves the whole floor, whose dimensions are $10^7 \times 10^7$ square millimeters.

Besides being a good programmer and ceramist, Vinko is also an excellent musician. Because of that, he knows that there are n points on the floor that are crucial for good acoustics in the hall. The hall's acoustics would improve significantly if a tile's corner is at one of the n points.



The left image shows the paving whose tiles have a diagonal of length 4. The point (2,4) is at the corner of a tile, and for it the acoustics are good, but for the points (4,3) and (5,1) are not.

The right image shows the paving whose tiles have a diagonal of length 2. The point (4,3) is at the corner of a tile, while points (2,4) and (5,1) are not.

Help Vinko determine for each of the n points in how many ways can he pave the floor (that is, how many different dimensions of the tiles can he choose) so that the i -th point is at a tile corner.

Input

The first line contains the integer n ($1 \leq n \leq 10^6$), the number of acoustic points.

The following n lines contain two integers x_i, y_i ($0 \leq x_i, y_i \leq 10^7$), which indicates that the i -th point is x_i millimeters away from the left wall, and y_i millimeters away from the lower wall of the hall.

Output

In the i -th of n lines print the number from the task statement for the i -th point.



Scoring

Subtask	Points	Constraints
1	15	$1 \leq n \leq 10\,000, 0 \leq x_i, y_i \leq 100$
2	55	$1 \leq n \leq 10\,000$
3	40	No additional constraints.

Examples

input

```
3
1 4
0 0
0 9
```

output

```
1
0
3
```

input

```
3
5 1
4 3
2 4
```

output

```
0
1
1
```

Clarification of the second example:

Images from the task statement correspond to the second example.



Task Neboderi

Domagoj is in the big city of London! Right now, there is a sequence of tall skyscrapers in front of him and he wants to take a photograph to remember the moment.

The sequence of skyscrapers can be represented as a sequence of n numbers $h_1, h_2, \dots, h_n$ where the number h_i represents the height of the i -th skyscraper. Domagoj will photograph a contiguous subsequence of skyscrapers. To capture more of the city's beauty, he wants to photograph at least k skyscrapers.



Domagoj has a strange sense of beauty of a photograph. He is very happy when there are tall skyscrapers in the photograph, but he is even happier when their heights have a large common divisor! If we label the heights of the contiguous skyscrapers on the photograph with $h_l, \dots, h_r$, and with g the greatest common divisor of the selected heights, then Domagoj defines the beauty of the photograph as $g \cdot (h_l + \dots + h_r)$.

Help Domagoj determine the beauty of the most beautiful photograph with at least k skyscrapers!

Input

The first line contains two integers n, k ($1 \leq k \leq n \leq 10^6$), the number of skyscrapers, and the number k .

The second line contains n integers $h_1, h_2, \dots, h_n$ ($1 \leq h_i \leq 10^6$), the heights of the skyscrapers, in order.

Output

Print a single line with the required number from the task.

Scoring

Subtask	Points	Constraints
1	11	$n, k \leq 100$
2	22	$n, k \leq 5000$
3	27	$h_i \leq 100$
4	18	$n, k \leq 5 \cdot 10^4$
5	32	No additional constraints.

Examples

input

6 2
2 1 4 4 4 2

output

48

input

4 1
7 3 9 4

output

81

Clarification of the first example:

Domagoj photographed skyscrapers (4, 4, 4), so the total beauty is $4 \cdot (4 + 4 + 4) = 48$

Clarification of the second example:

Domagoj photographed only the skyscraper (9), so the total beauty is $9 \cdot 9 = 81$.